



Starlink partners with airlines

SpaceX has started making deals with airlines to offer Starlink's satellite-based internet service to flyers. <https://digit.in/may22-01>



Microsoft Edge built-in VPN

In a support page, Microsoft revealed it's adding a free built-in VPN dubbed Edge Secure Network to Edge. <https://digit.in/may22-02>



TYPES OF OLED PANELS

QD-OLED vs OLED Ex vs OLED WBE vs OLED WBC TV panels

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It might come as a bit of a surprise to consumers that OLED panels for TVs haven't been updated very often. Almost all OLED TVs today source White OLED panels with White, Red, Green, and Blue sub-pixels from LG displays. White OLED was a revolutionary technology, but since 2016, we have seen only one major upgrade in White OLED TV panels which was marketed as OLED Evo in 2021. But if you are researching before buying an OLED TV, you will run into four different panel types of OLEDs

including QD-OLED, OLED WBE, OLED WBC and the latest OLED Ex.

QD-OLED is the latest entrant which will still take some time to mature, while White OLED panels are available in three varieties with OLED Ex being the most advanced offering. So, what's the difference between the four?

WHITE OLEDs: OLED EX VS OLED WBE VS OLED WBC

OLED WBC is the older panel that you get if you buy most of the last-gen OLED TVs like LG C1 or Sony A80J. Some of the 42-inch LG C2 models in 2022 were also found using the WBC panel, which is a clear indication that you might end up with an older WBC panel even for current-gen TVs based on the manufacturer you buy from and the market of purchase.

With the OLED WBE panel (popularly known as Evo panels), LG made two significant changes that helped increase the brightness by up

to 20 per cent over the OLED WBC. Each pixel of a White OLED screen (used on almost all OLED TVs) has a white, red, green, and blue sub-pixels. The OLED material only emits white light which is passed on as it is through the white sub-pixel and the other three colours – red, green, and blue – are achieved using a colour filter layer. The older WBC panels use two Blue OLED emitters with a yellow-green and red emitter in between to create white light. For the Evo or WBE panels, firstly, LG added an extra Green emitter layer between the two Blue OLED emitters (as shown in the image on the next page). This results in brighter and more efficient white light.

Secondly, the Hydrogen-based OLED material used in the blue emitters has been replaced with deuterium-based OLED material. The change in OLED material to deuterium has a significant impact and enhances the lifetime of OLED